Stat 266 – Unit 6 – Video Set 3 – Class Examples

Summary:

Method of Transformations (Bivariate)

Consider $U = h_1(X,Y)$ and $V = h_2(X,Y)$ to be two functions of the continuous random variables X and Y, with joint density $f_{X,Y}(x,y)$.

If the transformation from (x,y) to (u,v) is one-to-one for all $(x,\ y)$ (where $f_{X,Y}(x,y) > 0$) then the joint density of U and V is

$$f_{U,V}(u,v) = f_{X,Y}\left(h_1^{-1}(u,v), h_2^{-1}(u,v)\right)|J|$$

where

$$J = \det\begin{pmatrix} \dfrac{\partial h_1^{-1}}{\partial u} & \dfrac{\partial h_1^{-1}}{\partial v} \\ \dfrac{\partial h_2^{-1}}{\partial u} & \dfrac{\partial h_2^{-1}}{\partial v} \end{pmatrix} = \frac{\partial h_1^{-1}}{\partial u}\frac{\partial h_2^{-1}}{\partial v} - \frac{\partial h_1^{-1}}{\partial v}\frac{\partial h_2^{-1}}{\partial u} \quad \text{(called the Jacobian)}$$

- The inverse functions are $x = h_1^{-1}(u,v)$ and $y = h_2^{-1}(u,v)$.

Example A:

The life, (in hundreds of hours), of a type of battery running a particular machine follows an *Exponential* distribution with mean 1. All batteries are independent. Let X and Y denote the lives of two of these batteries. Suppose $U = X + Y$ and $V = \dfrac{X}{X+Y}$. Find the joint distribution of U and V. Given $f_{U,V}(u, v)$, find $f_U(u)$ and $f_V(v)$.

Since the $X, Y \sim$ Exponential (1) $\quad f_X(x) = e^{-x} \quad f_Y(y) = e^{-y}$.

X, Y are independent, joint distribution is $\begin{cases} e^{-(x+y)} & x>0,\ y>0 \\ 0 & \text{otherwise.} \end{cases}$

since $u = x+y \quad v = \dfrac{x}{x+y} \qquad u \cdot v = x. \qquad y = u - uv.$

$$J = \begin{bmatrix} \dfrac{\partial x}{\partial u} = v & \dfrac{\partial x}{\partial v} = u. \\ \dfrac{\partial y}{\partial u} = 1-v & \dfrac{\partial y}{\partial v} = -u \end{bmatrix} \qquad \det J = -uv - (1-v)u = u.$$

$$f_{UV}(u,v) = f_{X,Y}(uv,\ u-uv) \cdot |u|. = e^{-(u-uv+uv)} \cdot u = ue^{-u}.$$

since $0<x$, $0<y$ we have $0<x+y$, $\quad v = \dfrac{x}{x+y} \in (0,1).$

we want the marginal distribution of u, v

Then $f_u(u) = \int_0^1 u e^{-u} \, dv = u e^{-u}$ $\quad u \in [0, +\infty)$.

$$f_v(v) = \int_0^{+\infty} u e^{-u} \, du = -u e^{-u} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-u} \, du$$

$$= -e^{-u} \Big|_0^{+\infty} = 1.$$

the marginal distribution of v is uniform distribution

support on $(0, 1)$.

Example B:

Suppose the joint density of X and Y is

$$f(x,y)=\begin{cases}\dfrac{1}{x^2y^2} & ,1<x,1<y\\ 0 & ,\text{otherwise}\end{cases}$$

Consider the random variables $U=XY$, and $V=X/Y$.

a) What is the joint probability density of U and V?

b) What are the marginal probability densities of U and V?

$U=XY \quad V=X/Y \quad X=\sqrt{UV} \quad Y=\sqrt{\frac{U}{V}}$. (Note since $1<x$, $1<y$, the square root takes positive here.

$$J=\begin{bmatrix}\frac{\partial x}{\partial u}=\frac{V}{2\sqrt{uV}}, & \frac{\partial x}{\partial v}=\frac{u}{2\sqrt{uV}}\\ \frac{\partial y}{\partial u}=\frac{1}{2\sqrt{\frac{u}{v}}}\cdot\frac{1}{v}, & \frac{\partial y}{\partial v}=\frac{1}{2\sqrt{\frac{u}{v}}}(-u)\cdot\frac{1}{v^2}\end{bmatrix}.$$

$$|\det J|=\left|\frac{V}{2\sqrt{uV}}\cdot\frac{1}{2\sqrt{\frac{u}{v}}}\left(-\frac{u}{v^2}\right)-\frac{1}{2\sqrt{\frac{u}{v}}\cdot v}\cdot\frac{u}{2\sqrt{uV}}\right|$$

$$= \left| \frac{v\cdot(-u)}{4u\,v^2} - \frac{u}{4u\cdot v} \right| = \left| \frac{-1}{4v} - \frac{1}{4v} \right| = \frac{1}{2v}.$$

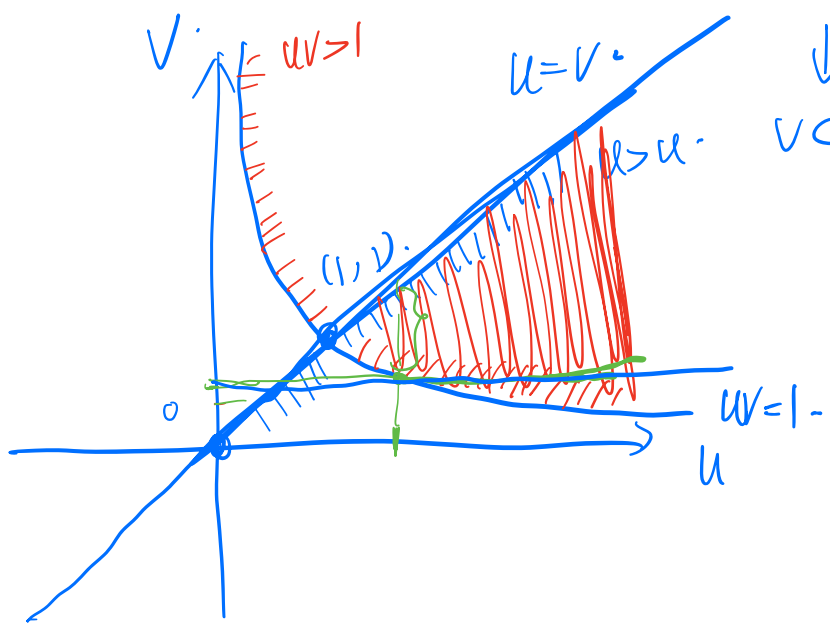
$$f_{U,V}(u,v) = f_{X,Y}\left(\sqrt{uv}, \sqrt{\frac{u}{v}}\right) = \frac{1}{(\sqrt{uv})^2 \left(\sqrt{\frac{u}{v}}\right)^2} \cdot \frac{1}{2v} = \frac{1}{2u^2 v}$$

$1 < x = \sqrt{uv}$ $\Downarrow$ $1 < uv$

$1 < y = \sqrt{\frac{u}{v}}$ $\Downarrow$ $1 < \frac{u}{v}$ $\Downarrow$ $v < u$.

$1 < u^2 \Rightarrow 1 < u$ $\frac{1}{u} < v < u$.

consider $uv = 1$ $v = u$.



$$f_U(u) = \int_{\frac{1}{u}}^{u} \frac{1}{2u^2 v}\, dv = \frac{1}{2u^2} \ln v \Big|_{\frac{1}{u}}^{u} = \frac{1}{2u^2}\left(\ln u - \ln \frac{1}{u}\right) = \frac{1}{u^2} \ln u.$$

$u > 1.$

when $v < 1.$ $u \in \left(\frac{1}{v}, +\infty\right).$

$$f_V(v) = \int_{\frac{1}{v}}^{+\infty} \frac{1}{2u^2 v}\, du = -\frac{1}{2u \cdot v} \Big|_{\frac{1}{v}}^{+\infty} = \frac{1}{2}$$

when $v \geq 1.$

$$f_V(v) = \int_{v}^{+\infty} \frac{1}{2u^2 v}\, du = -\frac{1}{2uv} \Big|_{v}^{+\infty} = \frac{1}{2v^2}$$

Example C:

Consider two random variables with joint density given as

$$f_{X,Y}(x,y)=\begin{cases}\dfrac{1}{\beta^2}e^{-y/\beta} & , \quad 0<x<y,\\ 0 \ , \quad else\end{cases}$$

Let $U=\dfrac{X}{Y}$. Find $f_U(u)$.

Solve the problem in two stages.

1. Change this problem to what we familiar with. Let's fix $Y=y_1$ is a constant. Then $U=\frac{X}{y_1}$

$X=y_1 u = h^{-1}(u)$.

$$f_U(u \mid Y=y) = f_{X,Y}(h^{-1}(u), y_1)\cdot\left|\frac{d(h^{-1}(u))}{du}\right| = \frac{1}{\beta^2}e^{-y_1/\beta}\cdot y_1$$

$$= \frac{1}{\beta^2}e^{-\frac{y_1}{\beta}}\cdot y_1$$

$$f_U(u) = \int_0^{+\infty}\frac{1}{\beta^2}e^{-\frac{y_1}{\beta}}\cdot y_1\,dx_1 = \left(-\frac{1}{\beta}y_1\right)\left(e^{-\frac{y_1}{\beta}}\right)\Big|_0^{+\infty} - \int_0^{+\infty}\left(-\frac{1}{\beta}\right)\cdot e^{-\frac{x_1}{\beta}}dx_1$$

$$= \frac{1}{\beta}\int_0^{+\infty} e^{-\frac{y_1}{\beta}}\,dy_1 = -e^{-\frac{y_1}{\beta}}\Big|_0^{+\infty} = 1.$$

Note that $u = \frac{x}{y}$ $\quad 0 < x < y \quad u \in [0, 1)$

So $f_U(u) = \begin{cases} 1 & u \in [0,1) \\ 0 & \text{otherwise.} \end{cases}$

Second method: Let $u = \frac{x}{y}$, $v = y$.

$x = u \cdot v \qquad y = v.$

$\frac{\partial x}{\partial u} = v \qquad \frac{\partial x}{\partial v} = u$

$\frac{\partial y}{\partial u} = 0 \qquad \frac{\partial y}{\partial v} = 1.$

Your Jacobian matrix: $\begin{bmatrix} v & u \\ 0 & 1 \end{bmatrix}.$

By theorem, we have the joint distribution of U, V is given by

$$f_{U,V}(u,v) = f_{X,Y}(uv, v) \cdot \left| \det \begin{bmatrix} v & u \\ 0 & 1 \end{bmatrix} \right|$$

$$= \begin{cases} \frac{1}{\beta^2} e^{-\frac{v}{\beta}} \cdot \underbrace{|v|}_{\text{since } v>0,\ |v| = v.} & 0<v,\ 0<u<1. \\ 0 & \text{otherwise.} \end{cases}$$

$$= \begin{cases} \frac{1}{\beta^2} \cdot v \cdot e^{-\frac{v}{\beta}} & 0<v,\ 0<u<1 \\ 0 & \text{otherwise} \end{cases}$$

we want the marginal distribution of u, thus $\int_{-\infty}^{+\infty} f_{U,V}(u,v)\, dv$

$$= \int_0^{+\infty} \frac{1}{\beta^2} v \cdot e^{-\frac{v}{\beta}}\, dv = 1.$$

Example D : The joint density of X and Y is given as

$$f(x,y)=\begin{cases}\frac{1}{2}, & 0<y<2, 0<x<1,\\ 0, & else\end{cases}$$

Find the probability density function of $U=Y-X$.

We use the second method to solve this:

Let $u=Y-X$, $v=X$. $Y=u+v$, $X=v$.

$$\text{Jacobian}=\begin{bmatrix}\frac{\partial x}{\partial u}=0 & \frac{\partial x}{\partial v}=1\\ \frac{\partial y}{\partial u}=1 & \frac{\partial y}{\partial v}=1\end{bmatrix} \qquad |\det(J)|=1.$$

$$f_{U,V}(u,v)=f_{X,Y}(Y-X, X)\cdot 1=\frac{1}{2}.$$

Note that $0<y<2$, $0<x<1$,

that implies $0<x=v<1$ $0<y=u+v<2$.

$0 < u+v < 2.$

$$f_{U,V}(u,v) = \begin{cases} \frac{1}{2} & -v < u < 2-v;\ 0 < v < 1. \\ 0 & \text{otherwise} \end{cases}$$

$$-v < u < 2-v \Rightarrow \begin{cases} -u < v \text{ and } v < 2-u. \\ 0 < v < 1. \end{cases}$$

$$f_U(u) = \int_{\max(0,-u)}^{\min(2-u,1)} \frac{1}{2}\, dv.$$

If $-1 < u < 0$, $\quad f_U(u) = \int_{-u}^{1} \frac{1}{2}\, dv = \frac{1}{2}(1+u).$

If $0 < u < 1$ $\quad f_U(u) = \int_0^1 \frac{1}{2}\, du = \frac{1}{2}.$

时 $1<u<2$ $\quad f_U(u) = \int_0^{2-u} \frac{1}{2}\,dx = \frac{1}{2}(2-u).$

0 otherwise.